MIL-C-0024643/52A(SH)
8 November 1985
USED IN LIEU OF
MIL-C-24643/52
28 September 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 300 VOLTS, TYPE LSTPNW (INCLUDING VARIATION LSTPNWA)

This military specification has been prepared by the Naval Sea Systems Command based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-C-24643. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-24643.

This cable can be manufactured in two variations: LSTPNW (unarmored) and LSTPNWA (armored).

(A) REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor, uncoated. AWG size 22, class K stranded.
Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Ethylene propylene rubber or cross-linked polyethylene insulation, 0.09 inch nominal thickness. Telephone identification code applied by method 3.
Fourth - Two conductors cabled together to form a pair. Maximum lay 2 inches.
Fifth - The required number of pairs (see table I) cabled with a lay not greater than 10 inches. Cabling sequence shall be consecutive, starting with pair no. 1, from center outward. Fillers shall be employed, as required, to form a firm, well-rounded assembly.
Sixth - Binder tape applied helically with overlap.

Construction variant for type LSTPNW

Seventh - Cross-linked polyolefin jacket. (See table I for thickness). Cable surface marking required.

(A) denotes changes.

FSC 6145

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Construction variant for type LSTPNWA

Seventh - Cross-linked polyolefin jacket. (See table I for thickness). Cable surface marking required.
Eighth - Braided metal armor and paint.

TABLE I. Details.

Military 1/ part no. M24643/52	Size 2/	Number of pairs	Cable jacket thickness nominal (inch)	Overall diameter (inches)				Cold bending mandrel diameter maximum (inches)
				Type LSTPNW		Type LSTPNWA		
				minimum	maximum	minimum	maximum	
-01UN	1-1/2	3/ 1-1/2	0.040	0.217	0.235	0.267	0.285	3.0
-02UN	3	3	.040	.287	.310	.337	.360	3.5
-03UN	5	5	.040	.338	.365	.388	.415	4.0
-04UN	10	10	.040	.435	.470	.485	.520	5.5
-05UN	15	15	.040	.490	.530	.540	.580	6.5
-06UN	20	20	.040	.532	.575	.582	.625	7.0
-07UN	30	30	.050	.629	.680	.679	.730	8.0
-08UN	40	40	.050	.708	.765	.758	.815	9.0

1/ Unarmored. When armor option is required, see MIL-C-24643 for configuration of part number.
2/ Type LSTPNW and LSTPNWA.
3/ Size 1-1/2 is composed of three conductors cabled to form a triad.

EXAMINATION AND TESTS:

	Requirements
(A) Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum.	17.94
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor.....	2000
Insulation resistance - megohms/1000 feet, minimum.....	1000
Conductor continuity.....	No failure
Group A:	
Visual and dimensional.....	No failure
Group B:	
Cold bending, cable - (See table I for mandrel diameter).....	No damage
Drip - 95 ± 1°C.....	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)..	35
Physicals (unaged)	
Insulation	
Ethylene propylene rubber	
Tensile strength - lbf/in^2, minimum.....	700
Elongation - percent, minimum.....	250

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EXAMINATION AND TESTS: (Continued)

	Requirements
Cross-linked polyethylene	
Tensile strength - lb/in^2, minimum.....	1800
Elongation - percent, minimum.....	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum.....	1300
Elongation - percent, minimum.....	160
Group C:	
Physicals (aged)	
Insulation	
Ethylene propylene rubber	
Air oven	
Tensile strength - percent of unaged, minimum.....	75
Elongation - percent of unaged, minimum.....	75
Cross-linked polyethylene	
Air oven	
Tensile strength - percent of unaged, minimum.....	80
Elongation - percent of unaged, minimum.....	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum.....	60
Elongation - percent of unaged, minimum..	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum.....	50
Elongation - percent, minimum.....	50
Shrinkage.....	No failure
Heat distortion - percent of unaged, maximum.....	30
Cable filler removability.....	No failure
Permanence of printing (jacket) type LSTPNW only - cycles, minimum.....	250
Armor - conformance to materials, construction and coverage, type LSTPNWA only.....	No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

Aging and compatibility (cable) (125 ± 5°C).....	No failure
Abrasion resistance (jacket/insulation).....	No failure
Acid gas equivalent - percent, maximum	
Jacket.....	2
Fillers.....	2
Insulation.....	18

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QUALIFICATION INSPECTION: (Continued)

	Requirements
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation (cable)	No failure
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	—
Insulation	10
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LSTPNW-1-1/2 through LSTPNW-10	2500
LSTPNW-15 and LSTPNW-20	1500
LSTPNW-30 and LSTPNW-40	1000

NOTE: Not for Air Force use.

Preparing activity:
Navy - SH
(Project 6145-N323-52)

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